

UNIT -IV

Management Information System (MIS): Classification and Quality of Information, Marketing models: Relational marketing, Sales force management, Logistic and production models: Supply chain optimization, Optimization models for logistics planning, Revenue management systems

1. Management Information System (MIS)

What is MIS?

A **Management Information System (MIS)** is a system used by businesses to collect, process, store, and distribute information needed to make better decisions. It helps managers track business performance and improve operations.

Example:

Imagine a **retail store**. The MIS tracks:

- **Sales data:** How many products were sold today?
- **Inventory data:** How much stock is left?
- **Employee performance:** How many sales did each employee make?

The store manager uses this data to:

- **Decide when to restock products.**
- **Identify the best-performing employees.**
- **Plan sales promotions** based on customer demand.

2. Classification of Information in MIS

Information in MIS can be classified into three types:

A) Strategic Information:

- Used for **long-term planning** and major business decisions.
- **Example:** Deciding whether to **open a new branch** in another city.

B) Tactical Information:

- Used for **short-term planning** and **day-to-day operations**.
- **Example:** Deciding **weekly discounts** on products based on sales data.

C) Operational Information:

- Used for **daily tasks** and managing **current operations**.
- **Example:** Tracking **today's sales** and **monitoring inventory levels**.

3. Quality of Information in MIS

For information to be **useful**, it needs to meet these **quality criteria**:

Criteria	Meaning	Example
Accuracy	Information must be correct and reliable .	If the MIS shows 50 units sold , it should reflect actual sales.
Timeliness	Information must be available when needed .	Daily sales reports help make quick decisions.
Relevance	Information should be important for the task .	Customer feedback is relevant when improving service quality .
Completeness	Information should be comprehensive .	Sales reports should cover all stores, not just one.
Consistency	Information should be uniform across reports .	If profit is ₹1L in one report, it shouldn't show ₹80K in another.

Marketing Models :

❓ Complexity of Marketing Decisions:

- Multiple objectives and countless alternative actions.
- Major choice options lead to a large number of possible decisions.

❓ **Role of Mathematical Models:**

- Help simplify and support marketing decisions.
- Use of large databases of sales transactions for accurate customer insights.

❓ **Two Key Topics in Marketing Intelligence:**

- **Predictive Models for Relational Marketing:**
 - Customize and strengthen company-customer relationships.
 - Different analysis streams and predictive models for various marketing problems.
 - Applicable to public administration-citizen relationships.
- **Salesforce Management:**
 - Decision-making in organizing a sales team.
 - Role of response functions in planning sales activities.
 - Use of optimization models for territory allocation and planning.

❓ **Real-World Application:**

- Business cases showcasing applied marketing models.
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Relational Marketing :

1. **Examples of Relational Marketing Initiatives:**
 - Insurance company: Identifying the best market segment for a new policy.
 - Mobile phone provider: Identifying customers likely to discontinue service (churn) and developing retention plans.
 - Bank: Selecting customers for a new savings management service offer.
2. **Common Features in These Examples:**
 - Companies have large databases describing customer purchasing behavior and service usage.
 - Goal: Extract useful, accurate knowledge to create targeted and effective marketing campaigns.
3. **Purpose of Relational Marketing Strategy:**

- Build, strengthen, and maintain long-term relationships with stakeholders (mainly customers).
- Involves: Analysis, planning, execution, and evaluation of marketing activities.
- 4. Emergence and Popularity:**
 - Became popular in the late 1990s for increasing customer satisfaction and achieving sustainable competitive advantage.
 - Led to cultural change in companies, focusing more on customers as key assets and sources of competitive advantage.
- 5. Adoption Across Industries:**
 - Initially adopted by financial and telecommunications service companies.
 - Later influenced consumer goods industries and manufacturing sectors like automotive and agricultural equipment.

Motivations and objectives :

1. Reasons for Spread of Relational Marketing Strategies:
 - Growth of large enterprises led to increased market complexity.
 - Shorter innovation-production-obsolescence cycles since the 1980s.
 - E-commerce and global comparisons of product features, prices, and reviews.
 - Decreased customer loyalty, especially in service industries.
 - Commoditization of products, where service quality becomes a key differentiator.
 - Automated sales data collection provides actionable marketing insights.
 - Advanced marketing data analysis techniques used by competitors.
2. Key Objectives of Relational Marketing:
 - Tailor offers for each customer segment through the right channel, timing, and pricing.
 - Use customer behavior data for targeted and effective marketing actions.
 - Convert occasional customer interactions into long-term relationships.
 - Achieve increased customer satisfaction and company profits.
3. Importance of Data and Tools:
 - Distinction between relational marketing strategy and CRM software.
 - Need for a company-wide vision focused on customer-centricity.
 - Integration of managerial culture, human resources, and data-driven tools.

- Use of business intelligence and data mining for enhanced customer information.
- 4. Challenges and Solutions:
 - Misconception that CRM software alone ensures relational marketing success.
 - High software costs lead to belief that relational marketing suits only large firms.
 - Essential components: well-designed data marts, analytical tools, and educated decision-makers.
- 5. Broader Relationship System:
 - Involves customers, employees, suppliers, and sales networks.
 - Challenges in accessing end-customer data due to intermediaries' reluctance.
 - Need for incentives and training to overcome data-sharing barriers.
- 6. Types of Customer Relationships:
 - High-value B2B customers require dedicated resources like key account managers.
 - Mass-market B2C customers engage through websites, call centers, and points of sale.
 - Data mining aids in segmenting and profiling fragmented customer interactions.
- 7. Cost vs. Revenue Balance:
 - Mass-marketing requires low-cost actions for low-revenue transactions.
 - Evolved customer relationships justify higher-cost, direct sales networks.
- 8. Customization and Cost Efficiency:
 - Mass-market companies must increase product/service customization.
 - Make-to-order companies must reduce costs and delivery times without sacrificing variety.
 - Data mining helps identify target segments for customized product offerings.



Figure 13.1 Decision-making options for a relational marketing strategy

An environment for relational marketing analysis :

1. Elements of Relational Marketing Analysis:

- **Information Infrastructure:**
 - Data warehouse: Integrates internal and external data sources.
 - Marketing data mart: Feeds business intelligence and data mining for customer profiling.
- **Use of Models:**
 - Pattern recognition and machine learning for customer segmentation.
 - Classification models create customer scoring systems for targeted marketing.
 - Example: Cross-selling offers directed only to high-probability customers to maximize campaign success.

2. Managing Marketing Campaign Cycles:

- Requires planning actions, communication channels, and resource use for each customer segment.
- Decision-making uses optimization models for campaign planning.
- Campaign cycle includes execution, data collection on results, and feeding new data into the marketing data mart.
- Control group: A similar set of customers who receive no action for assessing campaign effectiveness.

3. Types of Data Stored in a Data Mart:

- **Company Data Warehouse:** Customer demographics, administrative info, and transaction data.
- **Marketing Database:** Past campaigns, results, promotions, and customer value analysis.
- **Salesforce Database:** Contacts, calls, and sales conditions.

- **Contact Center Database:** Customer complaints, website navigation, forms, and email exchanges.

4. Data Analysis Process:

- **Exploration:** Analyze available customer data.
- **Inductive Learning Models:** Extract insights and rules to identify customer segments with similar behavior.
- **Customer Profiles:** Used to design and execute marketing campaigns.
- **New Data:** Generated from campaigns and reused in future analyses.

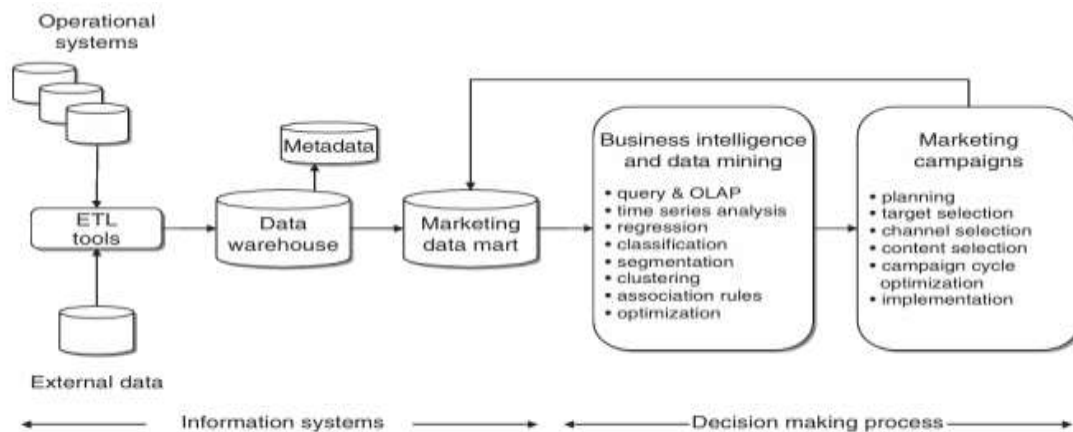


Figure 13.7 Components of an environment for relational marketing analysis

LifeTime Values :

1. Stages of Customer Lifetime:

- **Prospect Stage:** Potential customer who hasn't purchased yet.
 - Acquisition actions:
 - **Direct:** Calls, emails, sales agent talks.
 - **Indirect:** Ads, website notices.
 - Costs of acquisition cause an initial accumulated loss.
 - Example: A person browsing a bank's website without opening an account.

2. Customer Acquisition:

- Critical event: Prospect becomes a customer (e.g., service subscription, first purchase).
- Lost proposals: Rejected offers before becoming a customer.

3. **Maturity Stage:**

- Customer generates revenue, reducing initial loss and leading to cumulative profit.
- Marketing actions to extend profitability:
 - **Cross-selling:** Suggesting related products.
 - **Up-selling:** Offering higher-value products.
 - **Retention:** Keeping the customer loyal.
- Example: A bank offering a credit card to an existing savings account holder.

4. **End of Relationship:**

- **Voluntary:** Customer switches to a competitor.
- **Forced:** Non-compliance with payment terms.
- **Unintentional:** Change of residence.

5. **Main Tasks of Relational Marketing:**

- **Acquisition:** Identify high-potential customer segments by analyzing demographic and behavioral data.
- **Retention:** Predict and prevent customer churn by targeting at-risk high-value customers.
- **Cross-selling:** Identify customers likely to buy additional products.
- **Up-selling:** Promote higher-value products to existing customers.

6. **Role of Classification Models:**

- Used for accurate customer segmentation with a binary target class (e.g., will buy/won't buy).
- Attribute selection helps choose the most predictive features for meaningful rules.

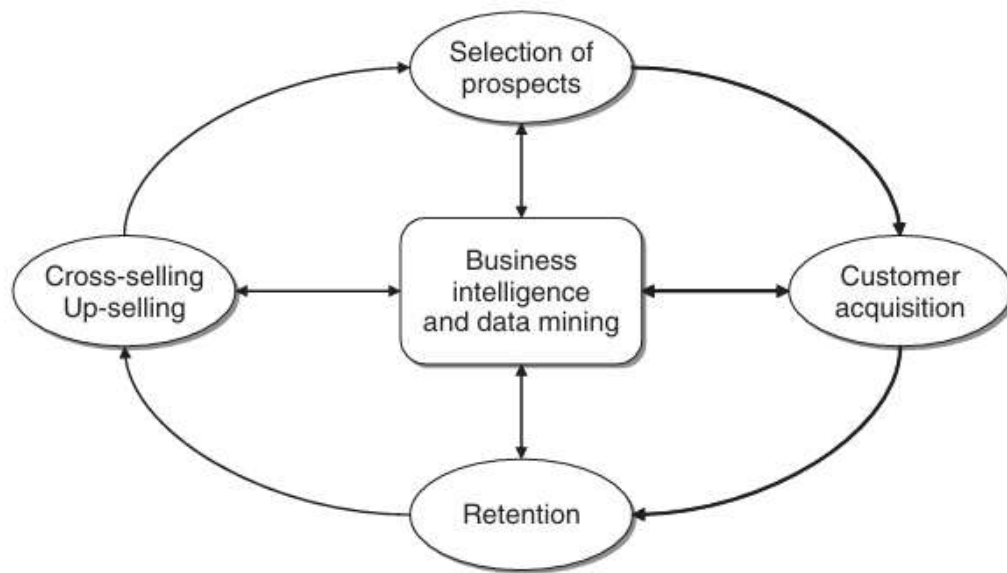


Figure 13.11 Main relational marketing tasks

The effect of latency in predictive models :

1. **Objective:** Develop a classification model for relational marketing considering time.
 - Example: A mobile provider predicts customer churn in November using data up to September.
2. **Data Availability:**
 - **Current Period (t):** October — cannot use this data for prediction as it's not yet available.
 - **Last Known Period (t-1):** September — used for target variable (identifies customers who churned).
 - **Two Periods Before (t-2):** August — not used for training to mirror real prediction conditions.
3. **Prediction Time Gap:**
 - Predict churn **h = 2 months in advance** (November prediction using data until September).
4. **Model Latency:**
 - **h-1 Periods (October):** Future data still unknown at prediction time — excluded from training.

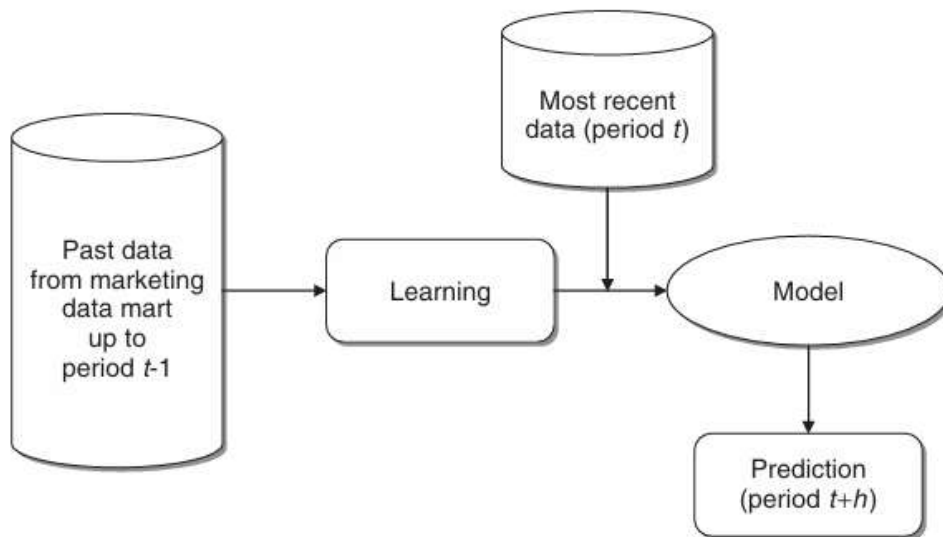


Figure 13.12 Development and application flowchart for a predictive model

Acquisition :

1. Importance of Acquisition:

- Even though retention is important, acquiring new customers is critical for company growth.

2. Identifying New Prospects:

- **Prospects:** Potential customers who:
 - Are unaware of the company's products/services.
 - Didn't have the characteristics to become customers before.
 - Were customers of competitors.
 - Are former customers who switched to competitors (more data available on them).

3. Targeting High-Potential Segments:

- Focus acquisition campaigns on segments with:
 - High potential profitability.
 - High probability of becoming customers.

4. Traditional Marketing Approach:

- Uses **top-down perspective** with:
 - Market polls.
 - Socio-demographic analyses.

5. Data-Driven Approach:

- Uses data from **data mart** (e.g., demographics, past contacts, competitor usage).
- Derives **classification rules** to identify promising customer profiles.
- This is a **binary classification problem**

Retention :

- 1. Market Saturation & Competition:**
 - Most products/services have matured, leading to market saturation.
 - Competitive conditions have become more intense.
- 2. Customer Base Expansion:**
 - Now relies more on **switching customers from competitors** (common in services like telecom, banking, insurance, and even manufacturing).
- 3. Attrition Analysis:**
 - Companies invest resources in understanding why customers **leave (churn) for competitors**.
- 4. Cost of Acquisition vs. Retention:**
 - Acquiring a new customer or winning back a lost one is **5 to 9 times costlier** than retaining an existing one.
 - Winning back a customer may **not always be successful** or profitable.
- 5. Challenges in Identifying Churn:**
 - Subscription services: **Clear signals** like formal withdrawal.
 - Other services: Need **indicators** (e.g., reduced purchases via loyalty cards or fewer bank transactions).
 - **Defining accurate threshold values** for churn is difficult.
- 6. Optimizing Retention Efforts:**
 - Focus on **high-value customers at risk of churning**.
 - Use **past churn data** to develop a **scoring system** predicting churn probability.
- 7. Churn Prediction & Marketing Action:**
 - Analyze **customer transactions** to identify key churn factors.
 - A **binary classification problem** (similar to previous models).
 - **Accurate targeting** reduces retention costs and increases effectiveness.

Cross-selling and up-selling :

- 1. Purpose of Data Mining Models:**

- Identify **market segments** with a **higher chance of buying additional products/services**.
- Example: A **bank offering insurance** may find customers interested in **life insurance policies**.
- 2. **Use of Data:**
 - **Demographic info** and **past transactions** from the **data mart** act as **explanatory attributes**.
 - Helps build a **classification model** to predict if a customer **accepts an offer or not** (binary variable).
- 3. **Cross-Selling:**
 - **Selling additional products/services** to an **existing customer** in a long-term relationship.
 - Classification models **identify customers** with a **high probability of accepting cross-sell offers**.
 - Example: A **bank customer** who **buys a life insurance policy** along with their **savings account**.
- 4. **Up-Selling:**
 - **Encouraging customers to upgrade** to a **higher-level product/service** with **more features** and **higher profitability**.
 - Example: **Upgrading from a standard credit card** to a **gold card** offering **extra services and benefits**.
- 5. **Binary Classification Problem:**
 - Both cross-selling and up-selling are **binary classification problems**.
 - Models use **training data on customer demographics and behavior** to **identify the best target segments**.

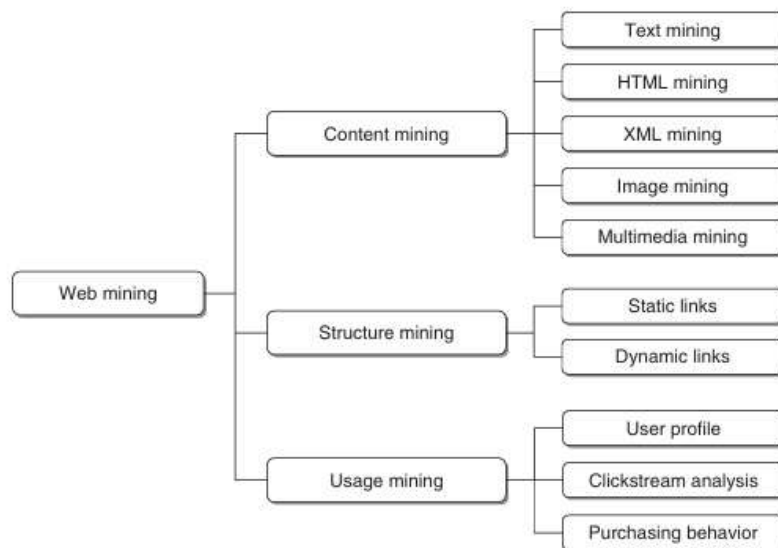
Market basket analysis

1. **Purpose of Market Basket Analysis:**
 - **Gain insights from customer purchases** to plan **better marketing actions**.
 - Mainly used in **retail and e-commerce**, also for **credit card purchases, phone services, banking products**, etc.
2. **Type of Data Used:**
 - **Purchase transactions** linked to **time dimension** (if customer is tracked via **loyalty card** or **invoice**).
 - Each transaction is a **list of purchased items**, called a **basket**.
3. **Association Rules (When Purchaser is Unknown):**

- **Identify item correlations** when **transactions aren't linked to a customer**.
- Helps in **decision-making** like:
 - **Item placement on shelves.**
 - **Store layout design.**
 - **Choosing items for promotions, flyers, or coupons.**
- 4. **Clustering Models:**
 - Groups **similar items together** using an **incidence matrix (X)**:
 - **Rows = transactions, Columns = items.**
- 5. **Customer-Identified Analysis:**
 - When **customers are tracked**, additional **time-based analyses** can be done:
 - **Sequential association rules** (analyzing **purchase patterns over time**).
 - **Time series analysis** for **predicting future purchases**.

Web Mining :

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1. Importance of Web Mining:

- Web is **crucial for company image and promotion**.
- **E-commerce sites are important sales channels**.
- **Web mining helps analyze visitor activities** on a website.

2. Types of Web Mining:

○ Content Mining:

- **Analyzes web page content** (texts, HTML, XML, images, multimedia).
- Used by **search engines** to **provide relevant links**.
- **Example: Text mining** for **classifying emails** in a support center.

○ Structure Mining:

- **Explores web's topological structure** using **links between pages**.
- **Creates graphs** (nodes = web pages, arcs = links).
- Identifies **popular sites, disconnected areas, maximal cliques**.
- **Local analysis:** Studies **site structure and user time on pages**.
- **Example:** If a **navigation page** takes too long, it may have a **problem**.

○ Usage Mining:

- **Analyzes user behavior and navigation paths** on a website.
- Uses **association rules** to find **correlations between visited pages**.
- Tracks users via **identification keys** or **cookies**.
- Uses **sequential association rules** and **time series models** for **time-based analysis**.
- Focuses on **clickstreams** (sequence of pages visited).
- **For e-commerce:** Tracks **purchase behavior**.

Salesforce management :

1. Importance of Salesforce:

- **Salesforce:** Group of people handling **sales activities**.
- **Critical for company profitability** and **relational marketing strategy**.

2. Types of Salesforce (Based on Activity):

○ Residential:

- **Sales at company sites** like **retail outlets, wholesale centers, and cash-and-carry shops**.

- **Example:** A customer buying groceries from a supermarket.
- **Mobile:**
 - **Company agents visit customers' homes or offices** to promote products and take orders.
 - **Mostly in B2B**, but also seen in **B2C**.
 - **Example:** A sales agent visiting a company to sell office supplies.
- **Telephone:**
 - **Sales through phone calls** to potential customers.
 - **Example:** A telecom company calling to offer a new internet plan.
- 3. **Challenges in Managing a Mobile Salesforce:**
 - **Designing the sales network.**
 - **Planning agents' activities.**
 - **Contact management** (managing customer interactions).
 - **Sales opportunity management** (tracking potential sales).
 - **Customer management** (handling customer relationships).
 - **Activity management** (tracking agents' work).
 - **Order management** (processing customer orders).
 - **Area and territory management** (assigning sales regions).
 - **Product and service configuration support** (customizing offers).
 - **Knowledge management** (sharing product and service info).
- 4. **Salesforce Automation (SFA):**
 - **Software tools help in managing operational activities.**
 - **Example:** Using a CRM tool to track customer orders and sales leads.

Decision processes in salesforce management :

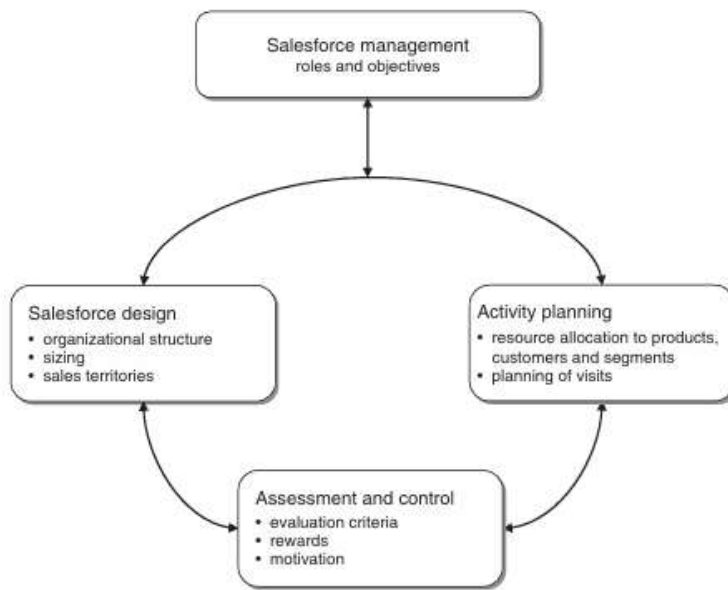


Figure 13.15 Decision processes in salesforce management

1. **Purpose of Salesforce Design and Management:**

- **Maximizes profitability.**
- **Increases sales effectiveness and resource efficiency.**
- Provides **better professional rewards** for sales agents.

2. **Decision-Making Categories:**

- **Design:** Creating and structuring the salesforce.
- **Planning:** Assigning resources and aligning with market goals.
- **Assessment:** Evaluating salesforce performance.

3. **Design Phase:** (During start-up or restructuring)

- **Market Segmentation:** Grouping customers using **data mining methods**.
- **Product & Service Classification:** Dividing offerings into **homogeneous classes**.

4. **Design Decisions:**

- **Organizational Structure:**
 - Grouping agents by **product, brand, geography, or market type**.
 - Deciding **level of specialization** based on **customer and product complexity**.
 - **Example:** Dividing agents into **separate teams for different product lines**.
- **Sizing:**
 - Deciding the **optimal number of agents**.

- Factors: **Number of customers, sales coverage, call time, travel time.**
- **Example:** Reducing the team may cut costs but **lower sales and revenue.**
- **Sales Territories:**
 - Grouping geographical areas and assigning them to agents.
 - Balancing **sales opportunities** across territories.
 - Considering **travel time, sales potential, and available agent time.**
 - **Example:** A sales agent assigned to **areas with similar sales potential.**

5. Planning Phase:

- Assigning **sales resources** (time, budget) to market segments.
- Considering **time for sales pitches, travel costs, and action effectiveness.**
- Involves **additional tasks** like product **demonstrations** and **future sales suggestions.**
- **Example:** An agent visiting a client to **explain product features** and **build future sales prospects.**

6. Assessment Phase:

- Measuring **effectiveness and efficiency** of sales agents.
- Designing **remuneration and incentive plans.**
- Defining **evaluation criteria** by considering each agent's **personal contribution.**
- Using **Data Envelopment Analysis (DEA)** for performance assessment.
- **Example:** Rewarding agents based on **individual sales performance** minus regional advantages.

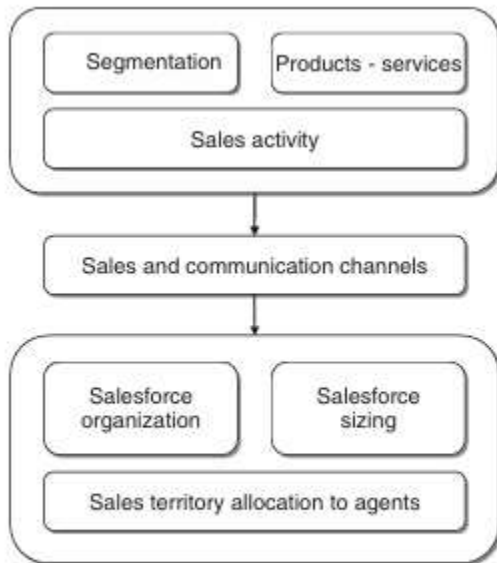


Figure 13.16 Salesforce design process

Models for salesforce management

1. **Purpose of Optimization Models:**
 - **Designing and planning the salesforce.**
 - **Educational use** to understand sales network optimization.
 - Real-world models are **more complex**, combining different features.
2. **Common Concepts for Models:**
 - **Sales Areas (J):** Region divided into **J geographical sales areas**.
 - **Territories:** Clusters of sales areas; each area belongs to **one territory**.
 - **Connection Property:** Every area in a territory is **reachable from others**.
3. **Time Span:**
 - Divided into **T equal intervals** (like weeks or months).
 - **Example:** $T = \{1, 2, \dots, T\}$ represents different time periods.
4. **Sales Agents:**
 - **Each territory has one agent.**
 - Agent's **residence area impacts travel time and cost**.
 - **Number of agents = Number of territories (I).**
5. **Customers and Segments:**
 - Each area has **customers or prospects** for sales and promotions.
 - Customers can be **grouped into segments** based on:
 - **Residence area.**

- **Value, development potential, purchasing behavior.**
- 6. **Market Entities (H):**
 - **H = {1, 2, ..., H}** represents **customers or customer segments**.
 - **D_j**: Set of **customers/segments in area j**.
- 7. **Products and Services (K):**
 - Each agent **promotes and sells K products/services**.
 - **Example: K = {1, 2, ..., K}** for different product lines.

Response Functions:

- **Meaning:** Response functions show how **customer sales respond to the efforts of sales agents**. It tells us **how much sales will increase** when agents make more calls, do product demos, or spend more time with customers.
- **Example:** If a sales agent **calls a customer twice a week** instead of once, sales might **increase by 20%**. The response function helps us **predict this change**.
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Sales Territory Design:

- **Meaning:** Dividing a **region into different sales areas** and assigning them to **different sales agents**. This helps manage customer coverage efficiently.
- **How it works:** We group nearby areas so that:
 - **Travel time is reduced.**
 - Each agent has a **balanced number of customers**.
 - **Sales opportunities are fairly distributed.**
- **Example:** Suppose Mumbai is divided into **5 sales areas** (like South, North, East, West, and Central). Each area is **assigned to one sales agent** so they focus only on that region.

Calls and Product Presentations Planning:

- **Meaning:** Deciding **how many calls and product demos** sales agents should make and **when** to do them. It helps **use their time effectively** and reach **more customers**.
- **What it includes:**
 - **Number of calls per customer** (weekly/monthly).
 - **Time spent on each call** (like 30 mins or 1 hour).

- **Which products to present** based on customer interest.
- **Example:** A sales agent plans to:
 - **Call 10 customers per week.**
 - **Spend 1 hour per meeting.**
 - **Present 3 different products** based on customer needs.

Logistic and production models :

1. **Combining Marketing and Data Mining:**
 - Using **relational marketing, business intelligence, and data mining** together helps **increase revenue and reduce costs**.
 - **Example:** A company analyzes customer data to **send targeted offers**, leading to **more sales with less advertising cost**.
2. **Supply Chain Management (SCM):**
 - **SCM focuses on logistics and production processes** — not just for one company but the **whole network of suppliers and manufacturers**.
 - **Example:** A car company's supply chain includes **parts suppliers, factories, and delivery services** working together efficiently.
3. **Integrated Planning for Logistics:**
 - Planning the **entire supply chain from one company's perspective** for **better coordination and cost savings**.
 - **Example:** A company plans **when and how much raw material to order** so they **avoid overstocking or running out of supplies**.
4. **Medium-Term Planning:**
 - Focuses on **important decisions for the supply chain over a few months or years** — not daily operations.
 - **Example:** Deciding **which warehouse to use** for **faster and cheaper delivery**.
5. **Optimization Models:**
 - **Mathematical models to improve logistics decisions** — like minimizing costs or delivery time.
 - **Example:** Using an optimization model to **find the shortest delivery route** for trucks.
6. **Real-Life Case Study:**
 - A **decision support system for supply chain optimization** was developed for a **food industry company**.

- **Example:** A bakery chain used software to **predict demand and manage ingredient supply**, reducing **waste and cost**.
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Supply chain optimization

1. Definition of Supply Chain:

- A **network of connected and dependent organizations** working together to **manage material and information flow** from suppliers to customers.
- **Example:** A smartphone company's supply chain includes **parts suppliers, assembly factories, and distributors**.

2. Integrated Planning and Operations:

- Combines and evaluates decisions and actions across different **subprocesses like procurement, processing, and distribution**.
- **Example:** A clothing brand plans **how much fabric to order, when to produce garments, and how to deliver them to stores efficiently**.

3. Efforts in Supply Chain Integration:

- Many companies, especially in the **consumer goods industry**, **integrate supply chain operations beyond their own company** — working with suppliers and distributors.
- **Example:** A food company collaborates with **farmers for raw materials and supermarkets for distribution**.

4. Goal of Integrated Logistics:

- **Minimize total costs**, including:
 - **Processing costs**
 - **Transportation costs** for procurement and distribution
 - **Inventory costs**
 - **Equipment costs**
- **Example:** A car manufacturer tries to **reduce storage costs** by **producing cars just in time for delivery**.

5. Holistic Cost Optimization:

- **Optimizing individual costs doesn't always minimize total cost** — a **system-wide approach** is necessary.
- **Example:** Cutting **transportation costs** by **using slower shipping** may **increase inventory costs** due to longer wait times.

6. Need for Logistic Optimization Models:

- **Medium-term planning and capacity analysis tools** are essential due to the **complexity of modern logistics**.
- **Example:** A tech company with **multiple production plants worldwide** needs models to plan **how to balance production across them**.

7. Features of Modern Logistic Systems:

- **Wide product range**
- **Multiple production plants and markets**
- **Large investments in automated technology**
- **Full usage of production capacity**
- **Short order-processing cycles**
- **Example:** An electronics brand with **high-tech factories** producing **various devices on tight schedules**.

8. Objective of Medium-Term Planning:

- Create a **logistic production plan** that **minimizes total costs**:
 - **Procurement costs**
 - **Processing costs**
 - **Storage costs**
 - **Distribution costs**
 - **Penalty costs** for failing to meet service levels
- **Example:** A furniture company **plans production** based on **raw material costs, factory capacity, and shipping times**.

9. Feasibility of Logistic Plans:

- Plans must meet **real-world constraints** like:
 - **Production capacity limits**
 - **Technological conditions**
 - **Material requirements**
 - **Logistic network structure**
 - **Minimum production batches**
 - **Decision-maker requirements**
- **Example:** A car company must **plan production** based on **factory capacity and parts availability**.

10. Power of Optimization Models:

- **Mathematical models** accurately **represent complex supply chain systems**:
 - **Capacity**
 - **Resources**
 - **Inventory**
 - **Batch sizes**
 - **Lead times**
 - **Logistic flows**
 - **Costs**

- **Example:** A pharmaceutical company uses optimization models to **balance medicine production and delivery costs**.

11. Advanced Decision Support Systems:

- **Modern information technology and optimization algorithms** enable **efficient development of decision support systems** for logistics planning.
- **Example:** A **retail chain** uses **software** to **forecast demand and plan inventory**, avoiding overstock or shortages.

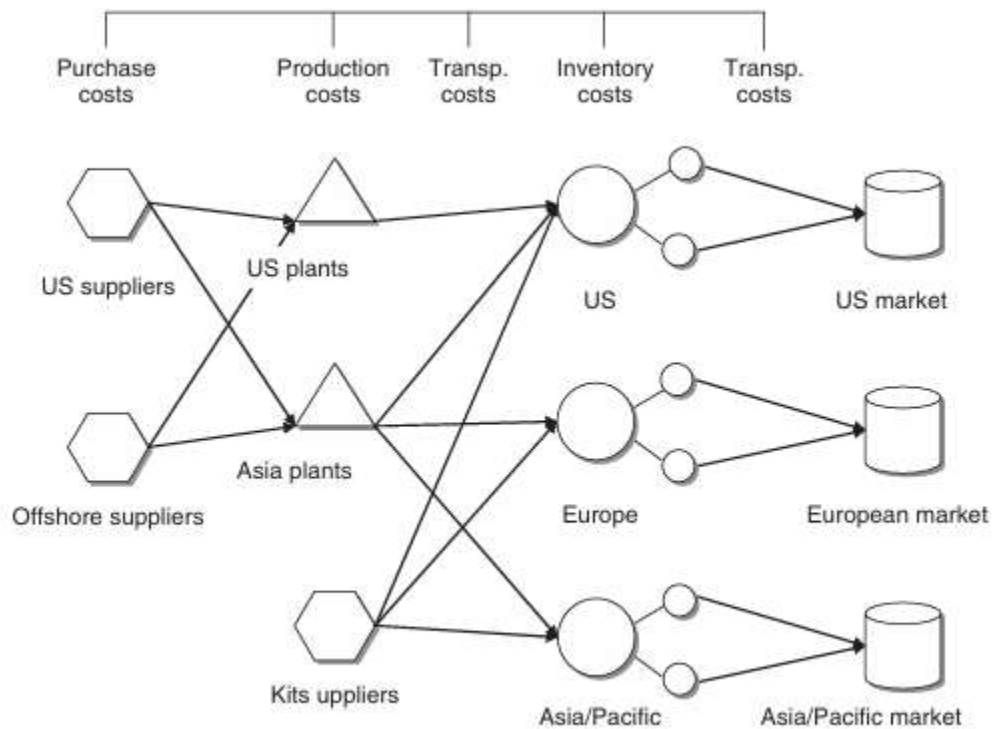


Figure 14.1 An example of global supply chain

Optimization models for logistics planning :

1. Purpose of Optimization Models:

- Used to **represent key features of logistic production systems**.
- **Example:** A factory uses an **optimization model** to decide **how many products to make each week**.

2. Simplified Illustration:

- Each model shows **one specific feature** of a logistic system for **simplicity**.
 - **Example:** A model may focus only on **manpower planning**, even though **real systems involve multiple factors**.
3. **Real-World Complexity:**
- **Real logistic systems combine multiple features**, making them **more complex**.
 - **Example:** A car company manages **production lines, material supply, and labor availability simultaneously**.
4. **Common Notation:**
- **Products (I):** Different products represented by the **index i** where $I = \{1, 2, \dots, I\}$.
 - **Example:** A bakery produces **bread, cakes, and cookies**, represented as $i = 1, 2, 3$.
5. **Planning Horizon (T):**
- **Time intervals for planning**, usually **weeks or months**, represented by $t = \{1, 2, \dots, T\}$.
 - **Example:** A clothing store plans its stock for each month over a **12-month horizon**.
6. **Critical Resources (R):**
- **Limited resources shared across products**, like **production lines, manpower, tools**, represented by $r = \{1, 2, \dots, R\}$.
 - **Example:** A toy factory shares **painting machines and assembly lines** for **different toys**.
7. **Simplification of Notation:**
- If **only one resource is relevant**, the **resource index (r)** is **omitted for simplicity**.
 - **Example:** When only **manpower availability matters**, other resources like **machines are ignored** in the model.

Tactical Planning:

- **What it is:** Medium-term planning (weeks or months) for managing production and supply chain efficiently.
- **Goal:** Balance **demand and supply** while controlling **costs and resources**.
- **Example:** A car company plans **how many cars to produce each month** based on **expected customer orders**.

Extra Capacity:

- **What it is:** Having **more production capacity** than the regular requirement, used during **high demand**.
- **Why it's needed:** To **avoid delays** when orders increase suddenly.
- **Example:** A **bakery hires temporary workers** during festival seasons to **bake more cakes**.

Multiple Resources:

- **What it is:** Using **different resources like machines, manpower, and materials** in the production process.
- **Importance:** **Efficient use** of all resources helps in **faster and cheaper production**.
- **Example:** A **furniture factory uses wood, cutting machines, and workers** together to **make chairs**.

Backlogging:

- **What it is:** **Delaying orders** when production **can't meet demand** immediately.
- **Why it happens:** Due to **limited resources or sudden high demand**.
- **Example:** An **electronics company delays laptop shipments** because **some parts are out of stock**.

Minimum Lots and Fixed Costs:

- **Minimum Lots:** The **smallest number of products** that must be produced in one batch.
- **Fixed Costs:** **Costs that don't change** even if production quantity changes (like **machine maintenance**).
- **Example:** A **clothing factory must produce at least 100 shirts** per batch because of **setup costs**.

Bill of Materials (BOM):

- **What it is:** A **detailed list of parts and materials** needed to make a product.
- **Why it's important:** Ensures **everything needed for production** is available.
- **Example:** A **pizza place's BOM includes dough, sauce, cheese, and toppings**.

Multiple Plants:

- **What it is:** A company having **more than one production facility** in **different locations**.
 - **Why it helps:** **Faster delivery, reduced transportation costs, and better resource sharing.**
 - **Example:** A car company has **factories in Mumbai, Pune, and Chennai** to serve different regions quickly.
-

Revenue management systems :

1. Definition:

- Revenue management helps **maximize profits** by balancing **demand and supply**.

2. Main Application Areas:

- Used in **marketing and logistics**.
- Popular in **air transport, tourism, hotels**.
- Now expanding to **manufacturing and distribution**.

3. Key Decisions in Revenue Management:

- **Pricing strategies** (how much to charge).
- **Sales promotions & discounts** (markdowns, special offers).
- **Product mix & allocation** (which products to sell where).
- **Multiple sales channels** (online, in-store, etc.).

4. Challenges in Implementation:

- **High costs** of setting up revenue management systems.
- **Uncertainty** about whether results will be profitable.
- **Complex decision-making process** in competitive markets.

5. Key Requirements for Success:

- **Optimized supply chain** (efficient production & delivery).
- **Deep customer knowledge** (understanding buying behavior).
- **Market segmentation** (grouping customers by preferences).
- **Data mining & business intelligence** (analyzing past sales data).

6. Use of Technology:

- **Data collection tools** (track customer purchases).
- **Analytical models** (predict demand & pricing strategies).

Example:

A **hotel** uses revenue management to:

- Increase **room prices** during peak holiday season.
- Offer **discounts** during off-season to fill empty rooms.
- Use **past booking data** to predict demand and adjust prices accordingly.

Decision processes in revenue management :

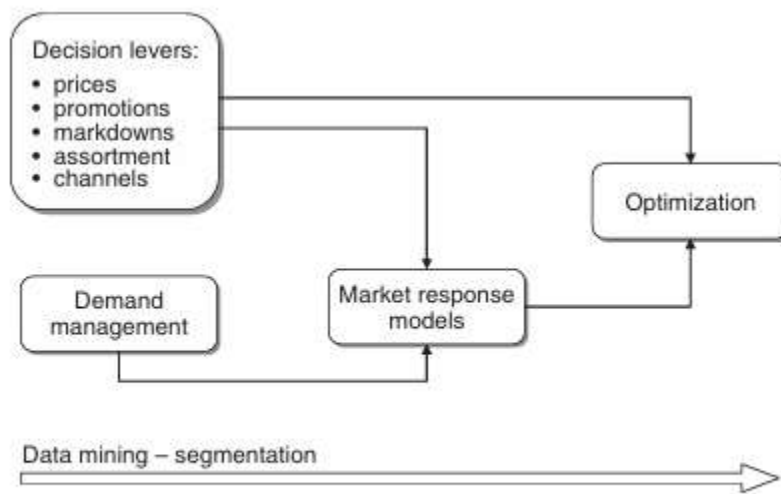


Figure 14.2 Decision processes in revenue management

1. **Definition:**
 - Uses **math models** to **predict customer behavior** and **set prices** to maximize **profits**.
2. **Purpose:**
 - **Match product supply with demand** by adjusting **prices, promotions, and assortments**.
3. **Focus:**
 - More on **demand management** than supply.
 - **Cost control** is more important than just increasing revenue.
4. **Industries Using Revenue Management:**
 - **Airlines, hotels, car rentals, theme parks, theaters, etc.**
5. **Common Characteristics of These Industries:**
 - **Perishable products** (e.g., unsold hotel rooms lose value).
 - **Low cost of additional sales** (e.g., filling an empty seat on a flight).

- **Dynamic pricing** (changing prices based on demand).
- 6. **Key Decisions in Revenue Management:**
 - **Market segmentation:** Dividing customers by **product, location, consumer type**, etc.
 - **Demand prediction:** Using **time series** and **regression models**.
 - **Optimal product mix:** Deciding which products to **sell where**.
 - **Market response analysis:** Understanding how customers **react to pricing** and competition.
 - **Pricing and markdowns:** Setting the **right price** and deciding **when to offer discounts**.
 - **Sales promotion planning:** Managing **offers** and **assessing their effectiveness**.
 - **Sales analysis:** Using data to **track performance** and **spot trends**.
 - **Stock management:** Deciding **when and how much to reorder**.
 - **Managing sales channels:** Coordinating **online, retail, and other channels**.
- 7. **Basic Principles of Revenue Management:**
 - **Micro-segmentation:** Using **data mining** and **business intelligence** to **understand customers**.
 - **Product value cycle:** Timing **availability and pricing** based on **demand changes**.
 - **Price-based approach:** Adjusting **prices instead of supply capacity** to balance **demand and supply**.
 - **Knowledge-based decisions:** Using **predictive models** to **forecast customer behavior**.
 - **Exploring new opportunities:** Continuously **looking for ways** to **increase revenue and profits**.
- 8. **Prerequisites for Success:**
 - **Data-driven culture:** Systematic **data collection, integration, and analysis**.
 - **Business intelligence:** Using **analytical models** for **market segmentation**.
 - **Supply chain integration:** Ensuring **efficient production** and **timely delivery**.
 - **Collaboration:** Strong **coordination between marketing and logistics teams**.
- 9. **Example:**

A hotel chain adjusts **room prices** based on **seasonal demand**, uses **promotions during low demand**, and **analyzes customer data** to offer **personalized discounts**.

Data Envelopment Analysis:

1. **Purpose:**
 - **Compare operating performance** of similar units like **companies, university departments, hospitals, banks, factories**, etc.
2. **Condition for Comparison:**
 - Units must be **homogeneous** (similar in nature).
3. **Performance Measurement Dimensions:**
 - **Quality indicators:** E.g., **rate of defective products** in a factory.
 - **Flexibility indicators:** E.g., **how quickly and cheaply a system adapts** to changes.
4. **Efficiency Measurement:**
 - DEA uses a **productivity indicator** to measure **efficiency**.
 - Efficiency is based on:
 - **Outputs:** Results achieved.
 - **Inputs:** Resources used to achieve results.
5. **Examples:**
 - **Bank branches:**
 - **Outputs:** **Number of active accounts, checks cashed, loans raised.**
 - **Inputs:** **Number of cashiers, managers, rooms used.**
 - **University departments:**
 - **Outputs:** **Active courses, scientific publications.**
 - **Inputs:** **Funding received, teaching costs, staff, office/lab availability.**

Efficiency measures :

1. **Decision Making Units (DMUs):**
 - DMUs are the units being compared (like companies, departments) with decision-making autonomy.
 - Set of DMUs: $N = \{1, 2, \dots, n\}$.
2. **Efficiency with Single Input and Single Output:**
 - Efficiency of $DMU_j = y_j / x_j$
 - y_j = Output produced by DMU_j
 - x_j = Input used by DMU_j

- *Example:* A factory uses 10 units of material (input) to produce 20 products (output). Efficiency = $20 / 10 = 2$.
- 3. **Efficiency with Multiple Inputs and Outputs:**
 - Efficiency is calculated using weighted sums:

$$\theta_j = (u_1 \cdot y_{1j} + u_2 \cdot y_{2j} + \dots + u_m \cdot y_{mj}) / (v_1 \cdot x_{1j} + v_2 \cdot x_{2j} + \dots + v_s \cdot x_{sj})$$
 - $u_1, u_2, \dots, u_m$ = Weights for outputs
 - $v_1, v_2, \dots, v_s$ = Weights for inputs
- 4. **Weight Sensitivity:**
 - Efficiency depends on the weights assigned to inputs and outputs.
 - Different weights can lead to different efficiency values, making it hard to find a single weight system acceptable for all DMUs.
- 5. **Best Weight System:**
 - DEA allows each DMU to choose the weight system that maximizes its own efficiency.
- 6. **Final Goal of DEA:**
 - Identify which DMUs are efficient in absolute terms through further analysis.

Efficient Frontier :

? Definition:

- The efficient frontier shows the maximum output possible from given inputs, or the minimum inputs needed for a specific output.
- It represents technically efficient operations.

? Empirical Derivation:

- It's built from observations of input/output combinations.
- Data envelopment analysis (DEA) is a method used to determine the efficient frontier.

? DEA vs. Regression:

- DEA is nonparametric (no pre-set shape), unlike regression.
- DEA identifies "best" performance, while regression shows "average" performance.

❓ Inefficiency Measurement:

- Units below the frontier are considered inefficient.
- Distance from the frontier measures inefficiency.

❓ Efficiency Ratios:

- Input-oriented efficiency (θ_I): Ideal input/actual input.
- Output-oriented efficiency (θ_O): Actual output/ideal output.

❓ Improvement Guidance:

- The frontier indicates how inefficient units can improve.
- It shows target input/output levels for efficiency.

❓ Graphical Representation:

- In a two-input, single-output scenario, inefficiency is shown by the distance from a unit to the frontier along a line from the origin.
- The production possibility set is the area under the frontier.

❓ Example of efficiency value:

- If you have a point A, and the efficient frontier is point P, and both are on a line from the origin, efficiency is the length of OP divided by the length of OA

CCR Model :

The CCR model, short for Charnes, Cooper, and Rhodes model, is a way to measure how efficiently something is working:

- **What it does:**
 - It helps compare the efficiency of different "things" (like companies, schools, or hospitals) by looking at their inputs (what they use) and outputs (what they produce).
 - It figures out which of these "things" are doing the best job of turning their inputs into outputs.

- **Key idea:**
 - It assumes that if you increase the inputs, the outputs will increase proportionally. This is called "constant returns to scale." Basically, if you double your effort, you should double your result.
- **How it helps:**
 - It can show which "things" are inefficient and how they can improve.
 - It helps to identify best practices by highlighting the most efficient performers.
- **In simpler terms:**
 - Imagine you're comparing different pizza restaurants. The CCR model helps you see which restaurant is making the most pizzas (outputs) with the least amount of dough, cheese, and toppings (inputs). It assumes that if a restaurant doubles its ingredients, it should be able to double its pizza output.

In essence, the CCR model is a tool for benchmarking and performance evaluation, focused on measuring efficiency under the assumption of a constant relationship between inputs and outputs.

Identification of good operating practices

- Think of it as finding the "best way" to do something.
- Like, if you're comparing different stores, it's finding what the most successful store does that makes it so good, so other stores can copy it.
- **Cross-efficiency analysis:**
 - Instead of just saying "this store is the best," it's like asking "how good would *other* stores say this store is?"
 - It gives a more balanced view of how well something is doing, because it gets opinions from everyone being compared.
- **Virtual inputs and virtual outputs:**
 - Sometimes, it's hard to measure things directly.

- Imagine trying to measure "customer satisfaction." You can't hold it in your hand. So, you might use a survey score as a "virtual output" to represent it.
- Virtual inputs are the same thing, but for the resources used.
- **Weight restrictions:**
 - When comparing things, we usually give importance to different factors.
 - Weight restrictions are like rules that say, "we can't give *too much* importance to this factor, or *too little* importance to that factor."
 - This stops the model from giving unrealistic results where one input or output is given total importance.